Many known numbers, one reading: a census of the constants touched by a two-mirror geometry, with a registered prediction for the dark-to-baryon ratio

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Abstract

Standard physics leaves many numbers free. This record lists the known numbers touched by the readings of the author's framework, Mathematical Materiality (MM), and asks which of them MM reads in a form that no other work we found gives. The work goes from the data to MM. A reading chosen after looking at the data is admitted if its choices are declared; only a fixed reading applied to data not yet seen is called a prediction. (1) One rule, two distant numbers. A single geometric factor, √3 per spatial direction between two mirror images, reads the cosmic dark-to-baryon ratio Ω_c/Ω_b = 3√3 = 5.196 and the galactic acceleration scale $a_0 = cH_0/6$. The two most recent CMB combinations, which share data, read 3√3 within 2σ: P-ACT-LB2 gives 5.199 and CMB-SPA + DESI DR2 gives 5.254 ± 0.037, with the correlations of the published chains. Planck 2018 alone is at 2.1σ. SPARC reads $cH_0/6$ within its systematic uncertainty. The reading was chosen knowing these values and its choices are declared: it is a coherent reading, not a derivation.

Many known numbers and a single rule: the list of the numbers touched by a two-mirror reading, and a prediction fixed in advance for the ratio between dark matter and ordinary matter

Student version of MM record 2.54 (DOI 10.5281/zenodo.23418156), written with an AI assistant from the archived text; the archived text remains on Zenodo.

The archived text is in English.

Version 2.54 of the Mathematical Materiality (MM) series, October 2026. The author is Ugo Lanciano, an independent researcher. Original text and materials: CC BY 4.0 licence. Original code: MIT licence. A licence says how other people can use the work.

The calculations were done with artificial intelligence (AI) assistants. Other AIs checked them. No human referee reviewed them. A referee is an expert who checks a work before a journal publishes it. The author's working files are in Italian.

This record continues MM 2.44. MM 2.44 checked whether the relation Ω_c/Ω_b = 3√3 is consistent with the data. The data were those of Planck and DESI DR2 (DOI 10.5281/zenodo.23329080). The record also refers to MM 2.46. MM 2.46 is about the galactic scale $cH_0/6$ in galaxy clusters (DOI 10.5281/zenodo.23340834). A DOI is a code that points to a document on the web forever. We will see what the symbols mean in section 1.

1. What this deposit is, and what it is not

A deposit is a work archived on the web, in a public archive. This section says briefly what the deposit contains. It also says what the deposit does not claim.

The question

In physics there are many important numbers. One example is how much dark matter there is in the universe, compared with ordinary matter. Standard physics leaves many of these numbers free. This means that it measures them, but it cannot say why they have exactly those values.

MM is a framework of ideas, that is, a general way of looking at nature. In MM, matter is a **reading** of **correlated copies**. Correlated copies are copies linked to each other. What happens to one says something about the others. Several MM readings use **mirror images**. Think of a drawing and its reflection in a mirror. They are two images of the same thing. In this text the word "reading" also has a simpler sense. It is a way of interpreting a number with the ideas of MM.

This deposit asks three questions, in this order:

1. Which known numbers are touched by the MM readings? And with which measurements are they compatible?
2. Which of these numbers does MM read in a form all its own? That is, in a form that no other work, among those found, gives?
3. How can that form be tested on data that do not exist yet?

Compatible means that the number and the measurement agree. The measurement errors are taken into account.

The work goes **from the data to MM**. First the measured data are looked at. Then an MM reading is sought. There are two levels:

- **Level 1, "coherent reading".** The reading is chosen after looking at the data. It is admitted only if its choices are declared, that is, written down clearly.
- **Level 2, prediction.** The reading is fixed first. Then it is applied to data that nobody has seen yet. Only this is called a prediction.

An everyday example. Saying the result of a match already played, when you know it, is not a prediction. Writing it down before the match, instead, is one.

The results

One rule, two distant numbers. A single geometric rule touches two numbers that are very far apart. Here is the rule. Between two mirror images there is a factor √3 for each direction of space. The symbol √3 means the square root of three. Space has three directions: length, width and height.

The first number concerns the whole universe. It is the ratio between dark matter and ordinary matter:

Ω_c/Ω_b = 3√3 = 5.196

Ω_c (omega c) is the part of the universe made of cold **dark matter**. Dark matter does not emit light. We recognise it only by its weight, that is, by how it attracts other things. The symbol Ω_b (omega b) stands for the part made of **baryons**. Baryons are protons and neutrons. They are the ordinary matter that atoms, stars and we ourselves are made of. In words, the formula says: dark matter is a little more than five times ordinary matter.

The second number concerns galaxies. It is an acceleration scale:

$a_0 = cH_0/6$

Here c is the speed of light. H_0 (H-zero) is the **Hubble constant**. It says how fast the universe is expanding today. a_0 (a-zero) is a very small acceleration. Think of the stars of a galaxy. When their acceleration drops below a_0, their motion changes. It no longer follows Newton's law applied to the visible matter alone. In words, the formula says: a_0 is the speed of light times the Hubble constant, divided by six.

What do the measurements say?

- The **CMB** is the cosmic microwave background radiation. It is the oldest light in the universe and it arrives from all directions of the sky. From the CMB one finds how much dark matter and how much ordinary matter there are.
- The measurements come from **combinations** of data. A combination is a set of different measurements put together. P-ACT-LB2 and CMB-SPA are the names of two combinations. Planck, ACT and SPT are telescopes

that observe the CMB. DESI is an instrument that measures the position of a very large number of galaxies. DR2 means "second data release".
- The two most recent CMB combinations read $3\sqrt{3}$ within 2σ. The symbol σ (sigma) stands for the typical error of a measurement. "Within 2σ" means this. The distance between the measurement and the number is less than two times the typical error.
- P-ACT-LB2 gives 5.199 ± 0.040. The sign $\pm$ (plus or minus) gives the error: the true value probably lies in that band.
- CMB-SPA + DESI DR2 gives 5.254 ± 0.037. This value is 1.6σ away from $3\sqrt{3}$.
- These errors use the correlations of the published **chains**. The chains are long lists of possible values, produced by the calculations of the collaborations. A **correlation** says how much two quantities are linked. If one is off upwards, the other tends to be off in a certain direction.
- The two combinations share part of the data.
- Planck 2018, alone, is at 2.1σ.
- SPARC reads $cH_0/6$ within its systematic uncertainty. SPARC is a catalogue of galaxies whose rotation speed is well known. The systematic uncertainty is the error that comes from the measuring method, not from chance. An example: a badly calibrated scale is always off in the same direction.
- The reading was chosen knowing these values. For this reason it is a coherent reading, not a derivation. A **derivation** is a calculation that obtains a number from principles, without looking at the measurement first.

The census. A census is an ordered list, like the list of the inhabitants of a town. Here it is the list of the known numbers touched by the MM readings.

- Section 2 is a selection from a working catalogue. The catalogue is still partial. From the old deposits, from 2.0 to 2.32, the numbers have not all been extracted yet.
- The census has 18 entries compatible with the measurements, or not excluded. Each entry holds under the conditions written in the entry itself. Then it has 7 entries with a future check.
- The entries are not 25 independent numbers. One entry rewrites another entry. One entry groups several structures. And some entries reappear among the future checks.
- The census also reports the unfavourable outcomes, that is, the results that went against a reading.
- Sometimes standard physics, or another author, already gives the same number or the same relation. The census says so. In those cases MM is compatible, but not distinctive: it says nothing different from the others.

The distinctive form. A distinctive form is a way of writing a number that, among the works found, only MM gives.

- Among the earlier works we identified, only one gives both relations. It is the **emergent gravity** of the physicist Verlinde (2016). In a theory of emergent gravity, gravity is not a basic force. It arises from something deeper.
- **Verlinde's two relations.** The first is $a_M = a_0/6$, with $a_0 = cH_0$. The second is $\Omega_D^2 = (4/3)\Omega_B$: it is his equation number 7.51. In words: the square of the dark-matter part is equal to four thirds of the ordinary-matter part.
- Verlinde presents the second relation as a conjectural extension. That is, as an extra hypothesis, not as a sure result. He himself calls it "a final fun comment", that is, a last comment made for fun.
- Here the second relation is taken as exact. And Ω_D, the dark-matter part in Verlinde's theory, is identified with Ω_c.
- **What it gives.** From these two choices a formula follows: $\omega_c/\omega_b = 2h/\sqrt{(3\omega_b)}$. Here h is the Hubble constant H_0, measured in kilometres per second per megaparsec and divided by a hundred: a pure number, without units of measurement. The quantities ω_b and ω_c (lower-case omega) are Ω_b and Ω_c multiplied by h^2. The ratio ω_c/ω_b is equal to the ratio Ω_c/Ω_b, because the factor h^2 cancels out. In words: in Verlinde's form the ratio depends on h and on ω_b.
- This ratio is equal to $3\sqrt{3}$ only when $\Omega_b = 4/81$. That is, only for a special value of h, which we call h* (h-star): $h^* = (9/2)\sqrt{\omega_b}$. In words: nine halves times the root of ω_b. With the values of ω_b used here, h* is about 0.675 (between 0.6730 and 0.6762).
- **The difference.** In MM the ratio is $3\sqrt{3}$ **for any value of h**. In Verlinde's form it is $3\sqrt{3}$ only when h is equal to h-star. Today h is close to h*. And the published data do not separate the two forms (section 4).
- An example to understand. Draw two lines on a sheet of paper: one flat and one going up. They cross at a point. Near the crossing they are almost at the same height, and they are hard to tell apart. Far from the crossing, instead, they separate. Here the flat line is the MM value, the same for every h. The rising line is Verlinde's value, which grows with h. The crossing is h*.

A registered prediction. Registered means written and fixed beforehand, with its date, so that it cannot be changed afterwards.

- Two future data sets may decide between the two forms. Or they may leave the comparison open.
- The first is the next **BAO** data release of DESI, the one that comes after DR2. BAO means "baryon acoustic

oscillations". It is a regular imprint in the position of galaxies. It works like a natural ruler for measuring the universe.
- The second is the first Simons Observatory parameters combined with DESI. Simons Observatory is a new observatory for the CMB.
- The rules for these data are fixed in advance. A program applies them (section 5).
- Only articles posted on the web after 16 October 2026, 02:22:54 UTC, are admitted. UTC is the reference time for the whole world.
- On that day an automatic search on arXiv found no article of this kind, among those already catalogued. The arXiv site is where physicists post their articles.

What it is not: what is not claimed

- **It is not a derivation.** Many entries of the census coincide with standard physics. Two examples are $\gamma = \beta = 1$ and the Tsirelson bound: we see them in section 2. Other entries coincide with relations proposed by other authors: Koide 1981, Brannen 2006, Rivero 2026, Verlinde 2016.
- **The strong coupling is not read.** The strong force holds the quarks together inside protons and neutrons. Its **coupling**, α_s (alpha s), is the number that says how intense it is. The reference value is given at the energy of the Z particle: it is written α_s(M_Z).
- In this work α_s(M_Z) enters as data, like the masses of the particles. It is not a result.
- The ratio **R** is built from measured α_s and α_2. α_2 is the coupling of the weak force. To build R, the **running** of the Standard Model is used. The running is the way in which a coupling changes with energy. The **Standard Model** is the theory of particles and of the forces between them, except gravity.
- Any reading that returns α_s through R is discarded by construction. It would be a closed loop: one would find again the data put in at the start.
- There are readings that give α_s as a result. Of these, some fell: the 28 registered counting mechanisms, and the reading with 16π. A counting mechanism is a way of obtaining a number by counting something.
- Two non-circular relations remain. Non-circular means that they do not find α_s again through R: they do not make the closed loop. Both were chosen after looking at the data.
- **First relation:** $\alpha_3 = (\alpha_2 + 2\alpha_Y)/3$ at the Planck scale. The **Planck scale** is an enormous energy, where gravity becomes as strong as the other forces. The symbol α_3 stands for the strong coupling. The symbol α_Y stands for the coupling of **hypercharge**, another charge of the particles. In words: the strong coupling is an average between the weak one, counted once, and the hypercharge one, counted twice.
- This relation still depends a little on α_s, through the running: by 1.9 %.
- Its **fixed point** is 0.11625 with the PDG 2025 inputs. It is 0.11666 with the PDG 2026 inputs. The fixed point is the value that, put into the calculation, comes out the same. The **PDG** is the group that collects the official measured values in particle physics.
- The comparison with the world average gives $z = -1.08$. The number z says by how many typical errors the value deviates. The minus sign means "below".
- With the ALPHA 2026 value as input, $z = -2.02$. ALPHA is a group that computes α_s on the **lattice**. The lattice is a computer calculation. In this calculation space and time are divided into a grid of points. This is a tension, not an exclusion. Tension means that the numbers move apart a bit too much. But not enough to discard the relation.
- This relation is submitted to a **registered prospective test**. It is a test written beforehand, on data that have yet to arrive. The data will be the next edition of the PDG and the next FLAG review. FLAG is a group that collects the lattice results. In this test the total error, σ_tot, counts. A compatibility counts as precise only if $\sigma_tot \leq 0.070$. That is, if the total error is at most 0.070.
- **Second relation:** Y/W = (F/W)^(3/2). Here the letters F, W and Y stand for the inverses of three couplings. F is for the strong force, W for the weak one, Y for hypercharge. So R = F/W. In words: the ratio Y/W is equal to the ratio F/W raised to the power three halves.
- This second relation is in tension with the ALPHA 2026 value: 3.3σ, counting only the errors of the input data. It was not tested further.
- **It is not a test of complete cosmologies.** A cosmology is a complete description of the universe and of its history. The parameters used here are those that the collaborations inferred within the **ΛCDM** model. The ΛCDM model is the standard model of the universe. It contains dark energy (Λ, lambda) and cold dark matter (CDM). The test compares only two relations among these parameters.
- **What a favourable outcome would mean.** It would be an **A-confirmation** on a parameter already used to choose the reading. "A-confirmation" is the name of one of the outcomes foreseen by the rules of section 5.
- A favourable outcome would support the MM reading of the ratio, in the domain examined. But it would not confirm MM as a whole. It would not prove that the ratio is independent of h for every value of h. And it would not prove that no other theory gives the same form.

2. The census: the list of the numbers

The census comes from a working catalogue written in Italian. The catalogue was registered on 16.10.2026. It collects:

- the deposits from 2.34 to 2.52;
- the book NUMERI;
- in part, the deposits from 2.0 to 2.32.

For each entry there is also the line "Also given by". It says where MM coincides with standard physics or with another author.

The number **R** appears in some entries. R is the ratio between the inverses of the strong and weak couplings, at the Planck scale: $R = (1/\alpha_3)/(1/\alpha_2)$. For the readings that use it, R is an **input**, that is, a starting datum. R has two values:

- 1.0768 ± 0.0014, with the running computed at two loops, as in MM 2.38;
- 1.0761 ± 0.0013, with the running computed at three loops (PDG 2025 inputs).

The **loops** are the degrees of precision of the calculation of the running. Three loops are more precise than two. The entry on the Higgs boson uses the first value. The entries on the CKM and on the neutron use the second. With the other value, each entry moves by less than its error.

Some numbers of this section can be redone starting from public data. Each of these was recomputed by a program. The check is described in section 4, under "Checks before publication".

2.1 Numbers compatible with the measurements, or not excluded, under the stated conditions

Here are the 18 entries. For each one we say four things:

- **read by MM:** the number that the MM reading gives;
- **measured:** what the measurements say;
- **also given by:** who else gives the same number;
- **conditions:** under which conditions the agreement holds.

The ratio between dark matter and ordinary matter.

- Read by MM: Ω_c/Ω_b = 3√3 = 5.196.
- Measured: P-ACT-LB2 gives 5.199 ± 0.040. CMB-SPA + DESI DR2 gives 5.254 ± 0.037, that is, 1.6σ away. Planck 2018 alone gives 5.364 ± 0.079, that is, 2.1σ. The errors use the correlations of the chains.
- Also given by: Verlinde 2016, but only if Ω_b = 4/81.
- Conditions: the two-mirror reading; the parameters of the ΛCDM model.

The acceleration scale of galaxies.

- Read by MM: $a_0 = cH_0/6 = (1.09\text{-}1.18)\cdot 10^{-10}$ m s⁻². In words: between 1.09 and 1.18 times 10^{-10} metres per second squared. 10^{-10} means "divided by a one followed by ten zeros". The metre per second squared is the unit of accelerations. It says by how much the speed grows in each second.
- Measured: SPARC gives $(1.20 \pm 0.02 \pm 0.24)\cdot 10^{-10}$. The first error, 0.02, comes from chance. The second, 0.24, is the systematic error.
- Also given by: Verlinde 2016 (a_M = $cH_0/6$). Sanders too noted $a_0 \approx cH_0/6$ as an empirical value (arXiv:1811.05260, 1912.00716). Empirical means found by looking at the data, without a theory. The sign ≈ means "approximately equal".
- Conditions: the same rule as the first entry; the agreement holds within the systematic uncertainty.

The part of ordinary matter in galaxy clusters.

- A galaxy cluster is a large group of galaxies held together by gravity. The **baryon fraction** is the part of the mass made of ordinary matter.
- Read by MM: 1/(1 + 3√3) = 0.161. In words: one divided by (one plus three root three). Think of it this way. For each part of ordinary matter there are 3√3 parts of dark matter. Then ordinary matter is one part out of 1 + 3√3.
- Measured: X-COP gives from 0.141 to 0.165, at the radius R500. X-COP is a programme that observes clusters. R500 is a reference radius used to measure clusters.
- Also given by: nobody.

- Conditions: it is the same ratio as the first entry, written in another way. A comparison cluster by cluster requires a model. The model must give the radius, the depletion, the stars and the gas. The depletion is how much less ordinary matter the cluster has compared with the average of the universe.

How a_0 changes over cosmic time.

- Read by MM: the shape of $a_0(z)$ is $a_0(z)/a_0(z_1) = E(z)/E(z_1)$, with $E = H/H_0$.
- Here z is the **redshift**. It measures how far back in time we look. The larger z is, the older the light we see. H is the speed of expansion of the universe at that time. $E = H/H_0$ compares it with today's. In words: a_0 changes over time as the speed of expansion changes.
- Measured: MUSE-DARK III, in four intervals of z, agrees within 1σ. MUSE-DARK III is a study of distant galaxies.
- Also given by: Milgrom 2015.
- Conditions: it is a registered prospective check, that is, fixed before the data of the check.

The Koide relation.

- The **charged leptons** are three particles: the electron, the muon and the tau. The Koide relation links their three masses. A certain ratio Q, computed from the masses, is exactly 2/3.
- Read by MM: Koide Q = 2/3 (charged leptons).
- Measured: 0.6666645 ± 0.0000051, that is, 0.4σ.
- Also given by: Koide 1981.
- Conditions: MM gives a reading of the relation, that is, a way of interpreting it. It does not give the number.

The down-type quarks, read through the mirror.

- The down-type quarks are three: down, strange and bottom. "Through the mirror" means using the inverse masses, 1/m, instead of the masses.
- Read by MM: the Koide relation with the inverse masses, Q(1/m).
- Measured: 0.6647 with the PDG masses, 0.6674 at the energy scale M_Z.
- Also given by: Rivero 2026.
- Conditions: the agreement holds within the uncertainties.

The sum of the neutrino masses.

- The **neutrino** is a very light particle. There are three types. The sum of their masses is written Σm_ν (sigma m nu). Their masses can be ordered in two ways, called normal and inverted. The eV (electronvolt) is a unit of energy and of mass used for particles.
- Read by MM: Σm_ν = 0.0593 eV, with normal ordering.
- Measured: there are several bounds in the ΛCDM model. The sign < means "less than".
- These allow it: < 0.064 (DESI DR2 + CMB) and < 0.071-0.077 (PDG 2025).
- These exclude it: < 0.048 (with SPT-3G) and < 0.053 (with Lyman-α). But together with it they exclude the whole normal ordering, which has a minimum of 0.0589. Lyman-α is a measurement made with the light absorbed by distant gas.
- In the w0waCDM model the bound is < 0.186. The w0waCDM model is a version of ΛCDM in which dark energy can change over time.
- Also given by: the same relation was proposed by Brannen in 2006. With the input data of that time it gave 0.0600 eV.
- Conditions: the value is allowed by the bounds that allow the normal ordering. A measured value near 0.06 eV would not single out this relation. That is, it would not be enough to say that this is exactly the right relation.

The mass of the W particle.

- The W particle carries the weak force. M_W is its mass. The GeV (gigaelectronvolt) is a unit of mass used for particles.
- Read by MM: M_W ≈ 80.379 GeV, from the pair of numbers (4, −2).
- Measured: 80.3692 ± 0.0133 GeV, that is, 0.7σ.
- Also given by: nobody.
- Conditions: the joint hypothesis (4, −2) is not validated. That is, the choice of those two numbers is not yet confirmed.

The Higgs boson at the Planck scale.

- The Higgs boson is the particle linked to the mechanism that gives mass to the other particles. The letter λ (lambda) stands for its **self-coupling**. It says how much the Higgs field acts on itself. M_P is the Planck mass.
- Read by MM: $\lambda(M_P) = -(R - 1)/(4\pi) = -0.0061$. In words: lambda is equal to minus (R minus one) divided by four pi.
- Measured: -0.0100 ± 0.0031, that is, 1.3σ.
- Also given by: $\lambda \approx 0$ was proposed by Froggatt–Nielsen 1995 and by Shaposhnikov–Wetterich 2010.
- Conditions: R is an input built from measured α_s and α_2. The measured λ(M_P) also uses α_s, in its running. So the two sides have correlated errors. The comparison requires their **covariance**, that is, the measure of how much the two errors go together.

The deficit of the first row of the CKM matrix.

- The CKM matrix says how easily a quark turns into another one, in the weak force. The numbers of the first row, squared and added up, should give exactly one. The **deficit** is how much is missing to reach one.
- Read by MM: $\delta^2/2 = 0.067$ %. In words: delta squared divided by two.
- Measured: 0.17 ± 0.07 %, that is, 1.5σ.
- Also given by: we found no earlier work on this value.
- Conditions: R is an input built from measured α_s and α_2. The target does not contain α_s. The measured deficit depends on V_us, one of the numbers of the first row.

The strong coupling with the levels reading.

- Read by MM: $\alpha_s(M_Z) = 0.1180$ with $c = 1/\sqrt{3}$. Here c is not the speed of light. It is a coefficient of the reading called the levels reading.
- Measured: 0.1180 ± 0.0009.
- Also given by: nobody.
- Conditions: it is a reading built on the data.

Two parameters of gravity: γ = β = 1.

- The letters γ (gamma) and β (beta) stand for two parameters of gravity. The first, γ, says how much mass curves space, for example by bending light. The second, β, says how much gravity adds to itself. In general relativity both are 1.
- Read by MM: $\gamma = \beta = 1$.
- Measured: the Cassini probe gives $\gamma - 1 = (2.1 \pm 2.3) \cdot 10^{-5}$. Cassini is a space probe that measured γ with radio signals.
- Also given by: general relativity.
- Conditions: only if R is a pure ratio of units.

The speed of gravitational waves.

- **Gravitational waves** are ripples of space and time.
- Read by MM: their speed is c, the speed of light.
- Measured: GW170817. It is the name of a gravitational wave that arrived together with the light of the same event.
- Also given by: general relativity.
- Conditions: it is as in general relativity (GR).

The lensing slip.

- **Lensing** is the bending of light caused by a mass, as in a lens. The slip η (eta) compares two ways in which gravity acts: on light and on slow matter. In general relativity it is 1.
- Read by MM: $\eta = 1$.
- Measured: KiDS gives 0.916, with an interval from 0.786 to 1.056. It is written [0.786; 1.056]. KiDS is a large survey of the sky.
- Also given by: general relativity.
- Conditions: it is as in general relativity.

The Tsirelson bound.

- In **quantum mechanics** two distant particles can have results linked to each other. There is a maximum value for how linked they can be: $2\sqrt{2}$. It is the Tsirelson bound.

- Read by MM: $2\sqrt{2}$.
- Measured: 2.82759 ± 0.00051.
- Also given by: quantum mechanics.
- Conditions: it is as in quantum mechanics, QM (MM 2.48).

Three-path interference.

- A particle can pass through several slits at once, like a wave. With three slits, quantum mechanics says that it is enough to add up the effects of the pairs of paths. The term with all three paths together, I_3, is zero.
- Read by MM: $I_3 = 0$.
- Measured: $< 10^{-2}$ (Sinha 2010).
- Also given by: quantum mechanics (Sorkin).
- Conditions: it is as in quantum mechanics.

How compact a star can be.

- The **compactness** of a star says how concentrated its mass is compared with its size. There is a maximum value.
- Read by MM: C = 2^(1/3) − 2^(−1/3) = 0.46622. In words: two to the one third, minus two to the minus one third.
- Measured: NICER gives a compatible result, but one that does not allow us to distinguish. NICER is an instrument that studies neutron stars.
- Also given by: the limit 4/9 for a perfect fluid (Buchdahl 1959). With the area variable the limit is 1/2 (Jacobson 2007). The area variable and the isotropic variable are two ways of measuring the radius of a star.
- Conditions: it is an ideal model, with uniform density, with the isotropic variable (MM 2.34). The mathematical, causal and stability limits are different.

Some structures of the particles.

- Read by MM: spin ½, two polarisations of the photon, eight gluons, three colours.
- **Spin** is a quantum property of particles that looks like a small rotation on themselves, but it is not a spinning ball. **Polarisation** is the direction in which light vibrates. The photon is the particle of light. **Gluons** are the particles that carry the strong force. The **colours** are the three charges of the strong force: they are not real colours.
- Measured: ½, 2, 8, 3.
- Also given by: D'Ariano–Perinotti; Wilson; Pati–Salam.
- Conditions: they are structures (MM 2.40, 2.52).

2.2 Numbers that can be checked in the future

These are the entries with a future check. For each one we say what the situation is today, and which check will be able to say more, and when.

$3\sqrt{3}$ for any value of h (against Verlinde).

- Today: the two forms are not separated (section 4).
- Possible check: DESI BAO after DR2, together with the CMB; Simons Observatory together with DESI (section 5).

The shape of $a_0(z)$.

- Today: it holds at 1σ.
- Possible check: new samples of galaxies at z = 1-2. The prediction was registered on 12.10.2026, in the working records.

Wide binaries follow Newton.

- **Wide binaries** are pairs of stars that orbit each other, very far apart. "Newtonian" means that they follow Newton's law of gravity.
- Today: with the Gaia DR3 data, in our analysis, $R \approx 1.25$. It is a moderate tension. Gaia is a satellite that measures the position and motion of stars. Here R stands for a ratio measured on the binaries. It is not the ratio between couplings of section 2.
- Possible check: Gaia DR4.

α_s(M_Z) from the relation $\alpha_3 = (\alpha_2 + 2\alpha_Y)/3$ at the Planck scale.

- The form of the relation was chosen after the data. It gives 0.11625 with the PDG 2025 inputs, and 0.11666 with those of the PDG 2026.
- Today: z = −1.08 against the world average. With the ALPHA 2026 lattice value as input, z = −2.02: it is a tension, not an exclusion. The relation is not circular.
- Possible check: the registered prospective test, in its version 3. It will use the next edition of the PDG and the next FLAG review. A compatibility is precise only if σ_tot ≤ 0.070. With today's other inputs, a lattice value of 0.1188 would give tension, not exclusion.

The beam/bottle difference of the neutron.

- The neutron, on its own, after a while turns into other particles. The time it lives is measured in two ways. In the *beam* method one observes a beam of neutrons in flight. In the *bottle* method the neutrons are closed in a container and one counts how many remain. The two methods give different values.
- Read by MM: the difference is δ/3 = 1.22 %. In words: delta divided by three.
- Today: the measured difference is 1.17 ± 0.23 %. But no rule chooses between the two possible readings. One gives −0.2σ, the other −5.7σ.
- Possible check: BL3 and UCNProBe, two new experiments on the neutron.

The deficit of the first CKM row.

- Today: 1.5σ.
- Possible check: the PIONEER experiment. Alone, with a precision of 0.05 %, it does not discriminate the predicted deficit. That is, it is not enough to distinguish it.

The sum of the neutrino masses and their ordering.

- Today: they are allowed by the bounds that allow the normal ordering.
- Possible check: Simons Observatory together with DESI; JUNO, a large neutrino detector.

The maximum mass of neutron stars.

- **Neutron stars** are remnants of stars, very small and very dense.
- Read by MM: the maximum mass is 5 % above that of general relativity.
- Today: it is **degenerate** with the equation of state. This means that the two effects cannot be separated. The equation of state is the law that links the pressure and the density of the matter of the star. By changing this law one can get the same effect.
- Possible check: NICER, eXTP.

2.3 Unfavourable outcomes: the readings that fell, were withdrawn or are still open

Here are the unfavourable outcomes. They are divided into four groups: readings that fell, readings withdrawn, open problems, coincidences not used.

Readings that fell.

- **The strong coupling obtained by counting.** 28 counting mechanisms had been registered. None hit the mark: 0 hits out of 28.
- **The "lit fraction".** A reading called the "lit fraction" gave 0.1671 for f_b. The quantity f_b is the part of ordinary matter in all matter. The value to compare with is 0.1571. The distance is 5.1σ from Planck 2018, counting the correlation of the chains. It is 7.6σ from CMB-SPA + DESI DR2.
- **Ω_Λ from rhythm × momentum.** Ω_Λ (omega lambda) is the part of the universe made of dark energy. The rhythm × momentum reading gave 0.483 or 0.820, against 0.698.
- **The law of the "side-by-side cubes".** It gives 2.43 at z = 1100.
- **"D = 3 selected".** It was the idea that the invariant at the two walls selected a space with three dimensions (D = 3). This reading fell.
- **NGC 2419.** It is a globular cluster, that is, a ball of old stars. The reading predicted a number ν (nu) between 2.75 and 3.06. Instead ≲ 1.1 is required. The sign ≲ means "less than or approximately equal to".

Withdrawn.

- **w ≥ −1 as an MM number.** The letter w stands for a number that describes dark energy.
- **The +43.9 nat on SPARC.** The nat is a unit that measures how much the data prefer one model to another.
- **The 30 pc cell.** The pc (parsec) is a unit of distance used for stars. The 30 pc cell is a **postulate** (erratum to MM 2.23). A postulate is a hypothesis taken as a starting point. An erratum is a document that corrects an earlier work. With this cell, the readings of the globular clusters and of the Coma cluster are withdrawn. Coma is a large cluster of galaxies. Saturn stays compatible with the cell.

Open problems or unfavourable comparisons still to be qualified. Still to be qualified means that we still have to understand well how much they weigh.

- **Galaxy clusters.** The acceleration scale in clusters is $g‡ = 2.02 \cdot 10^{-9}$ m s^{-2}. It is 16.7 ± 1.2 times the galaxy value. The two-mirror rule does not read it.
- **Diffuse dwarfs.** They are small galaxies of low density. The external field is the gravity of the nearby galaxies. With the external field, these dwarfs come out short by a factor 1.7-3.8 in σ. Here σ stands for how spread out the speeds of the stars are, not an error.
- **Palomar 14.** It is a globular cluster. The reading predicted from 0.79 to 1.11 km/s. The measurement gives 0.38 ± 0.12 km/s.
- **Dwarf spheroidals and ultra-faint dwarfs.** They are very small and very faint galaxies. They are not read without a law for each system.

Some outcomes come from extracts of the deposits from 2.0 to 2.32. They have to be reconstructed before using them again. Each unfavourable outcome concerns a reading, not the MM framework.

Coincidences noticed and not used. A coincidence is a number that comes out almost equal without a known reason. These coincidences were noticed, but they were not used:

- $2\pi - 1 \approx 5.28$;
- $R - 1 \approx 1/13$;
- $cH_0/(2e) \approx 1.20 \cdot 10^{-10}$ m s^{-2}. Here e is Euler's number, the base of natural logarithms;
- the mass of the top quark with 27π;
- Ω_b = 4/81, which makes 3√3 equal to Verlinde's value.

3. The two-mirror reading (level 1)

The reading has five choices. All of them are declared:

- **S1.** Between the two images there is a factor ℓ = √3 for each direction of space. The symbol ℓ is a cursive letter l.
- **S2.** Our matter keeps its mass. A quantity that contains h powers of length transforms as ℓ^h. Here h is only the number of powers: it is not the h of the Hubble constant. An example: an area contains length twice, because it is measured in metres times metres. Then it transforms as ℓ².
- **S3.** The content of the other image fills the volume. The integrals, that is, the sums made over the whole volume, grow as ℓ³. So Ω_c/Ω_b = ℓ³ = 3√3. In fact √3 times √3 times √3 makes 3√3.
- **S4.** The base acceleration is $cH_0/2$. It comes from the balance R_s/r = r/R_H. Here R_s is the Schwarzschild radius. It is the radius of a black hole with the same mass. The symbol R_H stands for the Hubble radius, a distance linked to the expansion of the universe. The letter r stands for the distance. In words: R_s is to r as r is to R_H.
- **S5.** If one reads along two directions, one obtains $a_0 = (cH_0/2) \cdot \ell^{-2} = cH_0/6$. In fact ℓ$^{-2}$ means divided by three, and one half divided by three makes one sixth.

The choices were made knowing the data. For this reason the reading is at level 1. With these choices the reading gives:

- two relations between numbers: 3√3, compared with P-ACT and with CMB-SPA + DESI, and $cH_0/6$, compared with SPARC;
- the cluster baryon fraction. It is a rewriting of the first relation. To compare it with X-COP a cluster model is needed;
- the shape of $a_0(z)$.

Without additions, the reading does not read:

- the normalisation of MUSE-DARK III, which is from 16 to 27 % higher. The normalisation is the overall level of the measured values;

- the groups of galaxies;
- the galaxy clusters;
- the dwarf galaxies;
- the globular clusters;
- the single galaxies.

Two AI reviewers checked this reading at level 1. Their outcome was "coherent with corrections". The corrections were applied.

4. The test on today's data: MM against Verlinde

In this section the two forms are compared with the data already published. One is the MM form, the other is Verlinde's.

Registration. The procedure of the test was registered before the calculation. The data, a deviation from the procedure and the program were also registered before the calculation. A deviation is a change with respect to the written procedure.

The method.

- **The correlations.** The published tables do not give the correlations between ω_b, ω_c and h. For this reason a grid of 32 correlation matrices was declared in advance. A **correlation matrix** is a table that says how linked each pair of quantities is. The grid tries many different possibilities. The published chains were downloaded afterwards. They give the actual correlations (see below, in the part on the checks).
- **The statistic.** For each relation the **χ^2** (chi-squared) is computed. The χ^2 measures how far a relation is from the data. If it is small, there is agreement. If it is large, there is disagreement. The errors are propagated linearly, that is, with a simple and direct method. Then the difference $\Delta\chi^2 = \chi^2_\text{Verlinde} - \chi^2_\text{MM}$ is computed. The symbol Δ (delta) means "difference". If $\Delta\chi^2$ is positive, MM does better. If it is negative, Verlinde does better.
- **The zones.** They were fixed in advance. The sign |…| means "the value without its sign".
- $|\Delta\chi^2| \leq 1$: no preference;
- from 1 to 4: a hint;
- > 4, that is, beyond 4: a preference.

combination	ω_c/ω_b measured	MM	Verlinde	$\Delta\chi^2$	zone
Planck 2018 (CMB only)	5.3643	5.1962	5.2035	−3.93 … −0.10	below or above the hint threshold depending on the correlations (always towards Verlinde)
P-ACT-LB2	5.1993	5.1962	5.2561	+1.02 … +2.35	hint towards MM
CMB-SPA + DESI DR2	5.2536	5.1962	5.2449	−2.80 … −1.85	hint towards Verlinde

Let us read the table row by row. The MM value is always the same, 5.1962. Verlinde's value changes from one row to the other, because it depends on h and on ω_b.

- **Planck 2018, CMB only.** Measured 5.3643. MM gives 5.1962, Verlinde 5.2035. $\Delta\chi^2$ goes from −3.93 to −0.10, depending on the correlations. It is below or above the hint threshold, depending on the correlations. But it is always towards Verlinde.
- **P-ACT-LB2.** Measured 5.1993. MM gives 5.1962, Verlinde 5.2561. $\Delta\chi^2$ goes from +1.02 to +2.35: a hint towards MM.
- **CMB-SPA + DESI DR2.** Measured 5.2536. MM gives 5.1962, Verlinde 5.2449. $\Delta\chi^2$ goes from −2.80 to −1.85: a hint towards Verlinde.

The outcome. With the registered numbers and grid, no combination reaches a preference. The two most recent combinations share data from Planck, ACT DR6 and DESI DR2. And they give hints in opposite directions. Today the test does not separate the two forms. The grade of this outcome is B*: it means that the data had already been seen.

The checks.

- **The Monte Carlo method.** The errors were also propagated with the Monte Carlo method. This method makes

very many random trials. The result agrees with the linear one within 0.01.

- **A more precise H_0 for P-ACT-LB2.** With $H_0 = 68.43 \pm 0.27$, taken from ACT DR6 (Table 5), the zone stays the same.
- **Wider correlation grids.**
- With 6,115 matrices, P-ACT-LB2 and CMB-SPA + DESI keep their hints. Planck stays undecided.
- With the 7,349 matrices of the prediction's grid, P-ACT-LB2 keeps its hint towards MM. CMB-SPA + DESI gives at least a hint towards Verlinde. Some matrices fall in the preference zone. With some matrices, MM ends up beyond 2σ.
- No grid gives a uniform preference, that is, one valid for all the matrices.
- **The correlations from the published chains.** The plan of this check was registered before downloading the chains. The program was registered before the calculation. Three chains are used:
- Planck 2018: plikHM TT,TE,EE + lowE + lensing. These are the names of the Planck data used;
- CMB-SPA + DESI DR2, in the corrected release of SPT-3G;
- for P-ACT-LB2 there is no public chain. So the P-ACT-LB chain with the DESI DR1 BAO is used. It serves only for its correlations.
- The means and the errors of the chains agree with the tables: within 1σ and within 20 %.

combination	ρ(ω_b, ω_c), ρ(ω_b, h), ρ(ω_c, h)	Δχ², registered grid	Δχ², chain correlations	zone
Planck 2018	−0.53, +0.71, −0.96 (outside the grid, mostly in ρ(ω_b, h))	−3.93 … −0.10	−2.15	hint towards Verlinde
P-ACT-LB2 (DR1 chain)	−0.26, +0.51, −0.93	+1.02 … +2.35	+1.09	hint towards MM
CMB-SPA + DESI DR2	−0.16, +0.46, −0.90	−2.80 … −1.85	−2.41	hint towards Verlinde

The symbol ρ (rho) is the **correlation coefficient**. It lies between −1 and +1. Near zero means weakly linked.

- **Planck 2018.** The correlations are −0.53, +0.71 and −0.96. They are outside the registered grid, especially ρ(ω_b, h). With the chains $\Delta\chi^2 = -2.15$. With the grid it went from −3.93 to −0.10. Zone: hint towards Verlinde.
- **P-ACT-LB2 (DR1 chain).** The correlations are −0.26, +0.51 and −0.93. With the chains $\Delta\chi^2 = +1.09$. With the grid it went from +1.02 to +2.35. Zone: hint towards MM.
- **CMB-SPA + DESI DR2.** The correlations are −0.16, +0.46 and −0.90. With the chains $\Delta\chi^2 = -2.41$. With the grid it went from −2.80 to −1.85. Zone: hint towards Verlinde.
- Now each combination falls in a single zone. Each $\Delta\chi^2$ lies inside the range of the registered test. This holds for Planck too, even though its correlations are outside the grid. No combination reaches a preference. So the outcome does not change: today the test does not separate the two forms.
- The **burn-in** is the first part of a chain, which is discarded. With other burn-in fractions the zones stay the same.

The checks before publication. They were done on 16.10.2026. Each program was registered before being run.

- **The census recomputed.** A program recomputed every number of section 2 that can be redone from public data. It used the PDG 2025, the CODATA 2022 and equation 61 of Buttazzo and colleagues (Buttazzo et al.). It also used the three-loop running of section 2 and the Planck chain. CODATA is the official collection of the physical constants.
- The program marked five lines as different (with the word DIFFERENT). They belong to two corrections already present in this text.
- First correction: the two values of R. It concerns three lines: R itself, and the CKM and neutron entries computed with R.
- Second correction: the error on f_b with the correlation of the chains. With this error the "lit fraction" is at 5.1σ instead of 6σ. It concerns two lines.
- The check also gave M_W at 0.74σ instead of 0.8σ. This was corrected in the table.
- It then gave an error of 0.0023 on the measured value of Y/W. The working catalogue had 0.0021.
- A third correction is the full list of the bounds on Σm_ν. It comes from the bounds themselves.
- All the other recomputed numbers agree with the text, at the digits written.
- **The earlier works.** We searched whether others had already written the same relations. The search was done in the full text of INSPIRE-HEP and in the abstracts of arXiv. INSPIRE-HEP is an archive of particle physics articles.
- No earlier statement of Ω_c/Ω_b = 3√3 was found. However, the search for the symbol √ is not reliable on INSPIRE. And ADS and Google Scholar, which are other archives of articles, were not consulted.
- For $a_0 = cH_0/6$, Sanders' empirical remark was found. It was added to the table.

- **The running.** The calculation of the three-loop running was compared with equation 61 of Buttazzo and colleagues (2013).
- The couplings at the Planck mass, M_Pl, agree within 0.00004. For λ, within 0.00015.
- The derivatives agree within 20 %. A derivative says how fast a number changes.
- The checks passed are 10 out of 10.
- The effect on the G2 statistic is 0.009. G2 is the name of the test on the relation $\alpha_3 = (\alpha_2 + 2\alpha_Y)/3$. The effect is below the rounding of the source.

5. The registered prediction

The prediction went through five versions, all registered. The one in force is version 5, registered at 02:22:54 UTC. It is the text of version 4, plus the corrections of the fourth verification round. The rules are applied by a program. If the text and the program disagree, the program prevails.

The targets. The targets are the future data on which the test will be made. Only articles posted on arXiv after the registration of version 5 are admitted.

- **DESI block.** It is the first DESI BAO release after DR2. More precisely, it is an article of the DESI collaboration. It is the article that combines with the CMB all the BAO tracers of that release. The **tracers** are the different kinds of galaxies and of other objects used to measure the BAO.
- **Simons Observatory (SO) block.** It is the first SO article with ΛCDM parameters. The parameters must combine the SO data with the DESI BAO.
- **The freezing.** Each block is "frozen" at a date, T_c. T_c is the date of the first arXiv version, plus 90 days, at 00:00 UTC. Frozen means that after that date what enters the test is no longer changed.
- **What enters:** the materials public at T_c. They are three things. The first is that version of the article. Then there are the corrections (errata) of the authors issued before T_c. Finally there are the chains available at T_c.
- **Errata issued later:** if an erratum is issued after T_c, a registered update is made. The update is placed next to the first result.

The data.

- **Admissible combinations.** They are those with the ΛCDM model, the CMB and the data that define the target. Nothing else.
- **The principal combination.** The program chooses it, in this order:
- among the admissible combinations that the authors indicate as reference (the *baseline*), if there are any;
- otherwise, among all the admissible combinations;
- if there is a tie, the one with the smallest error on $\Omega_c\, h^2$ wins. If there is still a tie, the smallest error on h wins. Then the one that comes first in the article wins.
- **The covariances.** They are taken from the published chains. If there are no chains, a fixed grid of 7,349 correlation matrices is used.

The rules. Each condition must hold on all the matrices of the principal combination. Several outcomes can hold together.

- **Exclusion.** A relation falls if $\chi^2 > 9$, that is, if the chi-squared exceeds nine.
- **Relative fall.** A relation also falls if all three of these conditions hold:
- its χ^2 is larger than 4;
- the other relation is preferred ($\Delta\chi^2$ beyond 4) and is compatible ($\chi^2 \le 4$);
- no single matrix of any other admissible combination gives $\Delta\chi^2$ beyond 1 in its favour.
- **A-confirmation of the MM reading.** It is a confirmation on a parameter already used to choose the reading. All three of these conditions are needed:
- MM is preferred ($\Delta\chi^2 > 4$);
- MM is compatible ($\chi^2 \le 4$);
- no single matrix of any other admissible combination gives $\Delta\chi^2 < -1$.
- **"7.51 favoured".** It is the symmetric outcome, in favour of Verlinde. It takes its name from his equation 7.51.
- **Otherwise** the test stays open.
- If a block has no admissible combination, the outcome is "not evaluable".
- If an input file is malformed, that is, written in the wrong way, the outcome is "invalid data". This is not a scientific outcome.

How the two blocks are combined.

- **Falls stand.** If a relation falls in one block, it stays fallen.
- **A confirmation in one block does not hold overall** in three cases. The first: the other block makes that reading fall. The second: the other block gives the opposite outcome. The third: in its principal combination, the other block shows a hint, at least a hint, or a preference in the opposite direction.
- **If a block is missing, or has not been evaluated,** the overall result stays provisional.
- **If some numbers were made public before the registration,** either confirmation drops to grade B*. And overall it stays at that grade.

If the MM reading falls, what falls is the MM reading of the ratio. MM does not fall.

When the test can separate the two forms. Some definitions are needed.

- $k = 2/\sqrt{(3\omega_b)} \approx 7.70$. In words: two divided by the root of three times ω_b.
- $h^* = (9/2)\sqrt{\omega_b}$, as in section 1.
- $R = \omega_c/\omega_b$ is the measured ratio. Here R is not the ratio between couplings of section 2. The symbol σ_R stands for the uncertainty of this ratio.
- P_V is Verlinde's value. The symbol σ_D stands for the uncertainty of the difference $R - P_V$.

There are three cases:

- If the measured ratio is equal to the MM one, then $\Delta\chi^2 = k^2(h - h^*)^2/\sigma_D^2$. In words: $\Delta\chi^2$ grows with the square of the distance between h and h-star.
- If the measured ratio is equal to Verlinde's, then $\Delta\chi^2 = -k^2(h - h^*)^2/\sigma_R^2$.
- If the measured ratio lies in between, $\Delta\chi^2$ can vanish or can change sign.

Today $h - h^*$ is +0.0010 for Planck, +0.0078 for P-ACT-LB2 and +0.0063 for CMB-SPA + DESI. Let us take the uncertainties of CMB-SPA + DESI and the grid of 32 matrices of section 4. To separate the two forms over that whole grid, in the MM case $|h - h^*| > 0.0133$ is needed. In the Verlinde case $|h - h^*| > 0.0108$ is needed. The MM case is the one in which the measured ratio is the MM one. The Verlinde case is the other one. With smaller uncertainties less is needed. This is a separation threshold, not a statistical power. The statistical power would be the probability that the test really manages to separate.

The verification. The rules were verified in four rounds by two AI reviewers. Each number of the reviewers was rechecked with code.

- **Round 1:** the numbers of the test were correct. The rules of the prediction had to be closed, that is, completed for every possible case.
- **Round 2:** a real error was found. In the program the confirmation could not be reached.
- **Round 3:** both reviewers found the decision core correct.
- **Round 4:** one reviewer gave "correct". The other found the core correct, and in addition six robustness defects. A robustness defect is a rare case in which the program could go wrong. These defects were closed in version 5. Now 71 edge-case tests pass. Among them are the exact counterexamples found by the reviewer.

What was registered, and when.

- **Before the calculation:** the procedure, the data and the program of the test.
- **After the calculation:** the outcome notes, the logs and the revised versions of the prediction, each with its own time. A log is the diary that a program writes while it works.
- **Before any target data:** the rules of the prediction.
- **The public date.** The times of the registrations were written on the author's computer, not in a public place. The public date will be the date on which this record is published. Whether this date comes before the target data will have to be checked at that time.

How the work was done

Who did the calculations. The calculations were done with AI assistants: Claude, with checks by Codex/GPT-6 and by Gemini. No human referee reviewed them.

The working names. In the working records this material is in the folder «MM_sviluppo_NUMERO» ("the distinctive number"). The working files are in Italian. The prediction was registered under the working names «previsione v1 … v5».

The registrations. A registered file has a SHA-256 fingerprint and a UTC time. The fingerprint is a code computed from the file: if the file changes even by one comma, the fingerprint changes. The registration lines (UTC time and SHA-256 of the registered files) are in `documenti/registri/`. They were registered locally, not publicly. The registrations cited in the text are:

- the census catalogue, `documenti/censimento/CATALOGO_NUMERI_v3.md`, registered on 16.10.2026 at 00:34:02 UTC;
- the procedure of the test of section 4, `documenti/test_oggi/PROCEDURA_LIVELLO2_K1.md`, at 01:10:49 UTC;
- the data, the deviation and the program of the same test, at 01:15:45 UTC (`DATI_E_DEVIAZIONE.md`, `codice/test_mm_verlinde.py`);
- version 5 of the prediction, at 02:22:54 UTC; the five versions are in `documenti/previsione/`;
- the plan of the check with the chains, before the download, and its program, before the calculation (`documenti/test_oggi/catene/`);
- the programs of the checks before publication, each one before being run (`documenti/verifiche_publish/`).

The arXiv search of section 1 is a query of the arXiv API of 16.10.2026, at 01:49 UTC. The answers are in `documenti/ineditezza/`.

The programs cited in the text.

- The value R = 1.0761 comes from the three-loop running. The running is computed by `codice/corsa3.py` (PDG 2025 inputs).
- The census is recomputed by `codice/ricalcolo_censimento_v2.py`. Its log marks the five lines DIFFERENT.
- The comparison of the running with eq. 61 of Buttazzo et al. (2013) is made by `codice/prova_corsa3_buttazzo_v2.py`.
- The rules of the prediction are applied by `codice/decidi_previsione_v5.py`.
- The relations of the force sector and the declaration that α_s is imported are in `documenti/forte/`. The G2 prospective test is in `documenti/forte/test_G2/`.
- The chain archives weigh 1.49 GB in all. They are not included. Their sizes and their SHA-256 are in SOURCES.md. Their weighted moments are in `dati/covarianze_catene.json`. From these, `codice/ricalcolo_catene.py` recomputes the last step.

The AI reviewers.

- The two-mirror reading (section 3) was checked by Codex/GPT-6 and by Gemini: "coherent with corrections".
- The four verification rounds of the prediction (section 5) are by Codex/GPT-6 and by Gemini. The reports are in `documenti/verifiche/`. In round 4 Gemini gave "correct". Codex found the six robustness defects. Its exact counterexamples are among the 71 edge-case tests.
- **External reading.** Before publication, Codex/GPT-6 read the corrections of that afternoon once: "ready with corrections", five points, all applied (`documenti/verifiche_publish/VERIFICA_ASTRA/`).

Related records. Before publication it was also checked that the two DOIs cited at the top lead to MM 2.44 and MM 2.46, as described.

What is in the package.

- `codice/`, the programs:
- test_mm_verlinde.py: the registered test;
- decidi_previsione_v5.py: the decision program;
- prove_decidi_v5.py: the 71 edge-case tests;
- ricontrollo_montecarlo.py, griglia_estesa.py, ricontrollo_numeri_astra.py: the rechecks;
- test_catene.py: reads the published chains. It requires pandas. The archives are listed in SOURCES.md;
- ricalcolo_catene.py: redoes the last step of test_catene.py from the stored moments;
- ricalcolo_censimento_v2.py: the census recomputed;
- prova_corsa3_buttazzo_v2.py: the running compared with Buttazzo et al.;
- run_all.py: reruns everything.
- `dati/`, the data:
- PROVA_FORMATO_DATI_2025_v4.json: the 2025 data in the input format of the program. It is a format check, not a test;
- covarianze_catene.json: the weighted means and the covariances of ω_b, ω_c, h from the chains.
- `risultati/`: the reference outputs of all the programs.
- `documenti/censimento/`: the Italian working catalogue v3.

- `documenti/test_oggi/`: procedure, data and deviation, outcome notes. Inside there is `catene/`, with the chain check, in Italian. It contains the plan, the list of files, the fingerprints, the log, the outcome note and the verification.
- `documenti/previsione/`: the five versions of the prediction.
- `documenti/verifiche/`: the reports of the four rounds.
- `documenti/ineditezza/`: the answers of the arXiv API of 16.10.2026, 01:49 UTC.
- `documenti/forte/`, in Italian. It contains three things:
- the dependency graph of the readings of the force sector, which pass through R;
- the declaration that α_s is imported;
- the check of the sensitivity of $\alpha_3 = (\alpha_2 + 2\alpha_Y)/3$ to α_s through the running.
- `documenti/forte/test_G2/`: the registered prospective test of $\alpha_3 = (\alpha_2 + 2\alpha_Y)/3$ on the next edition of the PDG and on the next FLAG review. It contains versions 1-3, the verification reports and the input data, in Italian. The program is codice/test_g2_v3.py. It uses the Standard Model running codice/corsa3.py (Buttazzo et al. 2013). There are 27 edge cases in codice/prove_g2_v3.py.
- `documenti/verifiche_publish/`: the four checks before publication. It contains the outcome note in Italian, the logs, the registrations and the fingerprints of the answers of INSPIRE, arXiv and Zenodo.
- `documenti/registri/`: the registration lines (UTC time and SHA-256) of the registered files.
- `SOURCES.md`: the measured values used, with their exact tables and equations.
- `IMPRONTE.sha256`: the fingerprints of all the files.

How to redo the calculations. The command is:

python codice/run_all.py

It reruns each program and compares its output with the reference files. It needs Python 3.9 or later, with numpy and scipy. It takes less than a minute. Before the deposit, a run from scratch on the extracted archive reproduced all ten outputs. The record of that run is kept with the upload files, outside the archive. Anyone can repeat it with the command written above.

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a full Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the reference. This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **A-confirmation.** The name of the favourable outcome of the prediction for the MM reading. It is a confirmation on a parameter already used to choose the reading.
- **a_0 (a-zero).** A very small acceleration scale that appears in the motion of stars in galaxies. MM reads it as $cH_0/6$.
- **BAO.** Baryon acoustic oscillations: a regular imprint in the position of galaxies, used as a natural ruler.
- **Baryons, ordinary matter.** Protons and neutrons: the matter that atoms, stars and we are made of. Their part in the universe is written Ω_b.
- **Census.** Here, the list of the known numbers touched by the MM readings.
- **Chains.** Long lists of possible values of the parameters, produced by the calculations of the collaborations. The correlations are read from them.
- **χ² (chi-squared) and Δχ².** The χ^2 measures how far a relation is from the data. The difference $\Delta\chi^2$ is the χ^2 of Verlinde minus that of MM. If it is positive, it goes towards MM. If it is negative, it goes towards Verlinde.
- **CMB.** The cosmic microwave background radiation: the oldest light in the universe.
- **Compatible.** A number is compatible with a measurement when they agree, taking the errors into account.
- **Correlated copies.** In MM, copies linked to each other. Matter is a reading of correlated copies.
- **Correlation, covariance.** How much two quantities, or their errors, go together.
- **Coupling (α).** The number that says how intense a force is. It is written α_s (or α_3) for the strong force, α_2 for the weak force, α_Y for hypercharge.
- **Dark matter.** Matter that does not emit light. We see it only by its weight. Its part in the universe is written Ω_c.
- **Derivation.** A calculation that obtains a number from principles, without looking at the measurement first. This work is not a derivation.
- **Grade B with an asterisk**, written B*. The grade of an outcome obtained on data already seen, or made public before the registration.
- **h-star**, written h*. The special value of h for which Verlinde's form gives exactly 3√3: about 0.675.
- **Hubble constant (H_0) and h.** H_0 says how fast the universe is expanding today. The letter h is H_0, measured in kilometres per second per megaparsec, divided by a hundred: a pure number, without units of measurement.
- **ΛCDM.** The standard model of the universe, with dark energy (Λ) and cold dark matter (CDM).

- **Lattice.** A computer calculation in which space and time are divided into a grid of points.
- **Level 1 and level 2.** Level 1: a reading chosen after seeing the data, with the choices declared ("coherent reading"). Level 2: a reading fixed beforehand and applied to data not yet seen (prediction).
- **Loops.** The degrees of precision of the calculation of the running. Three loops are more precise than two.
- **Mirror, mirror images.** Two images of the same thing, one the reflection of the other. The two-mirror reading puts a factor √3 for each direction of space between the two images.
- **MM.** Mathematical Materiality: the author's framework of ideas, in which matter is a reading of correlated copies.
- **ω_b and ω_c (lower-case omega).** Ω_b and Ω_c multiplied by h².
- **Outcome.** The result of a test. An unfavourable outcome makes a reading fall, not the MM framework.
- **Planck scale.** An enormous energy, where gravity becomes as strong as the other forces.
- **R.** In section 2, the ratio between the inverses of the strong and weak couplings at the Planck scale. In section 5, the measured ratio ω_c/ω_b. For the wide binaries, a ratio measured on the binaries.
- **Reading.** In MM, matter is a reading of correlated copies. In this text a reading is also a way of interpreting a number with the ideas of MM.
- **Registered prediction.** A rule written and fixed before seeing the data, with its date. So it cannot be changed afterwards.
- **Running.** The way in which a coupling changes with energy.
- **σ (sigma).** The typical error of a measurement. "At 2σ" means at a distance of two typical errors.
- **Tension.** A disagreement that is a bit too large, but not large enough to discard a relation.
- **Verlinde, emergent gravity.** A theory of 2016 in which gravity arises from something deeper. It is the only work found that gives both relations of MM, in a different form.
- **Working records.** The notebooks and the files in which the author notes down the work and the registrations.
- **Zones.** The bands fixed in advance for reading Δχ²: no preference, hint, preference.

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